

Optimization

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macroeconomics, demographic economics,
growth, public economics

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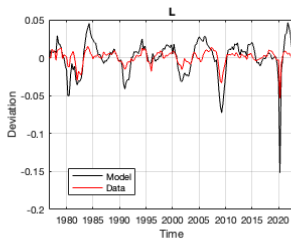
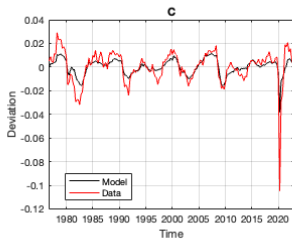
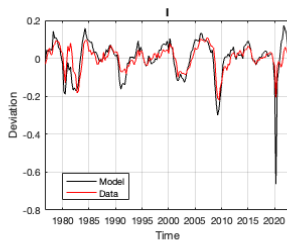
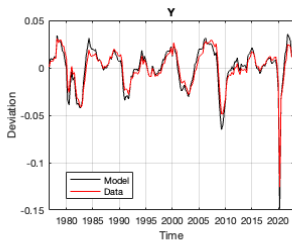
General equilibrium, choice, game theory,
network theory, public economics

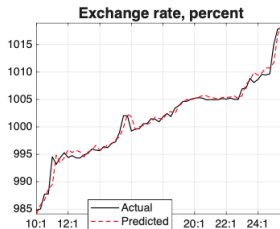
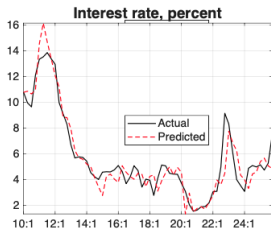
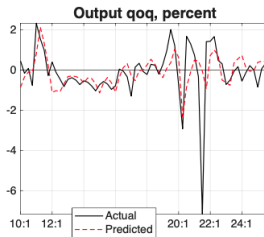
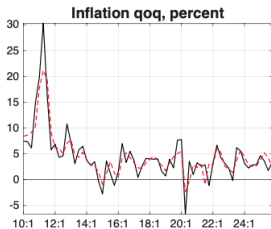
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Behaviors under scarce resources among competing uses.

- Macro (market economy, central bank problems)
- Micro (consumer problem, firm problems, strategic optimization)
- Econometrics (OLS, MLE, GMM)





Main topics

- ① Convex optimization
 - unconstrained
 - equality constraints
 - inequality constraints
- ② Linear programming

Grading

- Final exam: multiple choice
- Midterm: 2 small projects
- Attendance: missing 3 or more classes disqualifies you from the final exam
- Volunteering to answer questions, doing exercises in class, and submitting homework earns bonus points toward the midterm score.

- EMEA : Essential mathematics for economic analysis (Sydsaeter & Hammond et. al, 6e 2021)
- FMMA : Fundamental methods of mathematical economics (Chiang & Wainwright, 4e 2005)
- FMEA : Further mathematics for economic analysis (Sydsaeter & Hammond et. al, 2e 2008)
- ME : Mathematics for economists (Simon & Blume, 1994)
- NMO : Numerical methods and optimization (Butenko & Pardalos, 2014)
- ORAA : Operations Research: Applications and Algorithms (Winston, 2003)